

Internship Project Report: Deep Learning Image Classification using Transfer Learning

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Program Details

Program: Interspark AI/ML Internship Program

Domain: Artificial Intelligence & Machine Learning

Abstract

Image classification is one of the most important applications of computer vision and deep learning. This project develops a high-accuracy image classification model capable of distinguishing between cats and dogs using Transfer Learning with MobileNetV2.

The project demonstrates practical implementation of deep learning concepts including data preprocessing, data augmentation, transfer learning, model training, evaluation, and model persistence.

Objective

The primary objective of this project is to build and evaluate a deep learning image classification model using a pretrained neural network architecture and achieve high prediction accuracy on unseen image data.

Problem Statement

Traditional machine learning approaches require extensive feature engineering for image classification tasks. Deep learning models automatically learn hierarchical image features, making them highly effective for computer vision applications.

This project addresses the problem of binary image classification by identifying whether an input image belongs to a cat or dog category.

Dataset Description

Dataset: Cats vs Dogs

Source: TensorFlow Datasets

Total Images: 23,262

Classes:

- Cat
- Dog

The dataset contains real-world images with significant variation in scale, background, pose, and lighting conditions.

[Image Placeholder: Dataset Preview]

Methodology

Data Preprocessing

The following preprocessing operations were applied:

- Image resizing
- Pixel normalization
- Dataset batching
- Dataset shuffling

Data Augmentation

To improve generalization:

- Random Horizontal Flip
 - Random Rotation
 - Random Zoom
-

Model Architecture

The model uses MobileNetV2 pretrained on ImageNet as a feature extractor.

Additional layers:

- Global Average Pooling
- Dropout Layer
- Dense Output Layer

Transfer Learning was applied by freezing pretrained layers and training only the custom classification head.

[Image Placeholder: Model Architecture Summary]

Training Configuration

Optimizer: Adam

Loss Function: Binary Crossentropy

Epochs: 5

Batch Size: 32

Input Resolution: 224 × 224

Results

Performance Metrics

Training Accuracy: 94.76%

Validation Accuracy: 98.28%

Validation Loss: 0.0460

[Image Placeholder: Training Accuracy Curve]

[Image Placeholder: Training Loss Curve]

[Image Placeholder: Evaluation Metrics]

[Image Placeholder: Sample Prediction Output]

The results demonstrate strong predictive capability and excellent generalization performance.

Discussion

The use of Transfer Learning significantly reduced training time while improving classification performance. MobileNetV2 successfully extracted meaningful image features, resulting in high validation accuracy despite limited training epochs.

Data augmentation further improved model robustness by exposing the network to diverse image transformations.

Conclusion

This project successfully implemented a deep learning image classification system using TensorFlow and MobileNetV2. The model achieved over 98% validation accuracy, demonstrating the effectiveness of transfer learning for image classification tasks.

The project provided hands-on experience in computer vision, deep learning model development, evaluation, and deployment preparation.

Future Scope

- Fine-tuning pretrained layers
 - Multi-class classification
 - Streamlit-based deployment
 - TensorFlow Lite optimization
 - Mobile application integration
 - Cloud deployment
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GitHub Repository

(Add your repository URL here)

Declaration

This project was developed independently as part of the Interspark AI/ML Internship Program for educational and internship submission purposes.

Dog



Dog



Dog



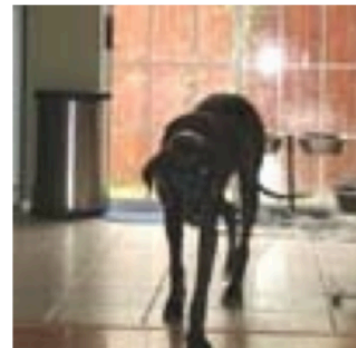
Cat



Dog



Dog



Cat



Cat



Dog



(32, 224, 224, 3)
(32,)

146/146 ————— 32s 65ms/step - accuracy: 0.9826 - loss: 0.0452
Test Accuracy: 0.9826
Test Loss: 0.0452

Model: "sequential_2"

Layer (type)	Output Shape	Param #
sequential_1 (Sequential)	?	0 (unbuilt)
mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	?	0
dropout (Dropout)	?	0
dense (Dense)	?	0 (unbuilt)

Total params: 2,257,984 (8.61 MB)

Trainable params: 0 (0.00 B)

Non-trainable params: 2,257,984 (8.61 MB)

